

Appendix B — Full Annotated Dry-Run Trace

B.1 What this trace is, and what it is not

This walkthrough follows one complete, real run of ``app/agent_query.py`` — the query “What is an adjustable speed drive and how does it control motor speed?”, captured in full in ``Runs/RAG Dry Run (3).txt``, dated 2026-05-08 — from the moment the question is typed to the moment the final answer prints. Every quoted line below is verbatim from that file; nothing is reconstructed or averaged across runs.

Common pitfall — This trace predates Phoenix and LangSmith — by design, not by omission

This particular run has no Phoenix or LangSmith span IDs anywhere in it, and that is expected rather than a gap in the log: it is dated 2026-05-08, and ADR-063/ADR-064 (Decisions.md) adopted Phoenix and LangSmith instrumentation on 2026-07-09 — two months later — and scoped that instrumentation explicitly to ``app_workflow/``, the later LangGraph rewrite, never to ``app/``, the pipeline this trace ran on. A from-scratch dry run of ``app/`` today would still show no span IDs; the two live in structurally different codebases within this project, not just different months. Section B.8 shows what a real Phoenix/LangSmith span ID actually looks like in this project's logs — sourced from a separate ``app_workflow/`` run, clearly labeled as such, rather than spliced into this trace as if it belonged to the same run.

B.2 Phase 1 — the question arrives, and context-size telemetry begins

The session banner, the question, and the orchestrator's own receipt of it, verbatim:

```
(RAG-work)      PS      C:\Users\LOQ\Desktop\Projects\RAG-work\app>      python
.\agent_query.py
Warning: You are sending unauthenticated requests to the HF Hub. Please set
a HF_TOKEN to enable higher rate limits and faster downloads.
Model 'all-MiniLM-L6-v2' loaded on cuda. Embedding dimension: 384
GPU: NVIDIA GeForce RTX 5050 Laptop GPU
Vector          store          initialized:          collection='documents',
path='../data/vector_store'
Existing documents in collection: 237
[SelfLearner] learned_qa collection has 0 entries.

=== Agentic RAG -- Self-Learning Query Interface ===
Self-learning triggers every 5 successful interactions.

Commands: 'bad' to flag last answer | 'stats' for learning stats | 'learn'
to force distillation | 'quit' to exit

Your question: What is an adjustable speed drive and how does it control
motor speed?

=====
STEP 1 -- USER QUERY RECEIVED
=====
Input : "What is an adjustable speed drive and how does it control motor
speed?"
Length: 70 chars
=====
```

There is no separate query-variant-generation step in this build — that becomes its own LangGraph node only in the later `app\_workflow/` rewrite. Here, the first LLM turn does both jobs at once. The complete message payload sent to the LLM — including the system prompt exactly as the CLI's own debug print renders it, truncated mid-sentence at “PROCESS (f...” in the original log itself, not by this appendix — is:

```
=====
FULL MESSAGE SENT TO LLM  [Iteration 1 / max 6]
=====

[0] role=system
    content: You are a research assistant. Answer ONLY from retrieved
chunks -- never from memory.

HARD LIMITS (enforced by the system -- do not test them):
- retrieve_documents: max 5 calls total. Stop at 3 if results are poor.
- Never repeat a query already tried. Never generate two queries with
identical words.
- If the knowledge base has no answer after 3 tries, say: "The knowledge
base does not contain information about [topic]."
- Never fabricate facts.

TOOLS:
- retrieve_documents(query, top_k): searches the knowledge base
- check_answer_quality(answer): validates your answer is grounded

PROCESS (f...

[1] role=user
    content: What is an adjustable speed drive and how does it control
motor speed?

Tool schemas available: ['retrieve_documents', 'check_answer_quality']
=====
```

Immediately before that payload is sent, the first context-size telemetry block fires — the human-readable form and its greppable `[CTXSIZE]` twin, back to back:

```
[CONTEXT SIZE @ iter 1]
  messages in payload      : 2
  total content chars      : 1,959
  ~tokens (chars/4 est.)  : 489
  actual prompt tokens     : 820   (from API)
  cumulative prompt tok    : 820
[CTXSIZE] iter=1 msgs=2 chars=1959 prompt_tokens=820 cum_prompt_tokens=820
```

The char/4 heuristic (489 tokens) undershoots the API's real count (820) by 40% on this very first call — a genuine, sourced reason the pipeline logs both numbers rather than trusting the estimate alone. The LLM's complete response requests three retrieval calls in a single turn, each with its own tool-call ID:

```
=====
LLM RESPONSE  [Iteration 1]
=====

Tokens this call : 820 prompt + 578 completion = 1398 total
content (0 chars):

tool_calls requested (3):
  -> name : retrieve_documents
      id   : h8qzjxhdv
      args : {
        "query": "adjustable speed drive",
        "top_k": 5
      }
  -> name : retrieve_documents
      id   : zar94qpyb
      args : {
        "query": "motor speed control",
        "top_k": 5
      }
  -> name : retrieve_documents
      id   : 4exprkj3t
      args : {
        "query": "adjustable speed drive motor control",
        "top_k": 5
      }
=====
```

B.3 Phase 2, call 1 of 3 — "adjustable speed drive"

Each tool call gets its own full embedding-to-search trace. The book's promise of an “embedding log” means exactly this — the real model name, the real output shape, and the first components of the real vector, not a description of what an embedding is:

```
-----  
STEP: QUERY -> EMBEDDING  
-----  
Query       : "adjustable speed drive"  
Model       : all-MiniLM-L6-v2  
Shape       : (384,)  
First 8 vals : [0.011542, 0.063031, 0.005661, 0.013145, -0.092976,  
0.013261, -0.07855, 0.003341]  
L2 norm     : 1.000000 (1.0 = fully normalised)  
-----  
  
Querying vector store -- top_k=5, score_threshold=0.5  
  
Merged & ranked 5 doc(s) after dedup/threshold filter:  
Doc 0 -- score: 0.8260 [..\data\Adjustable Speed Drive.txt]  
Doc 1 -- score: 0.8153 [..\data\Adjustable Speed Drive.txt]  
Doc 2 -- score: 0.8066 [..\data\Adjustable Speed Drive.txt]  
Doc 3 -- score: 0.7982 [..\data\Adjustable Speed Drive.txt]  
Doc 4 -- score: 0.7902 [..\data\Adjustable Speed Drive.txt]
```

The complete unfiltered text of all five retrieved chunks, before any relevance filtering:

```
-----  
RETRIEVED (unfiltered) -- retrieve_documents("adjustable speed drive")  
5 chunk(s) | 1946 chars total -- pending relevance check  
-----
```

```
[Source: ..\data\Adjustable Speed Drive.txt, score: 0.826]  
Adjustable Speed Drives provide precise control over motor speed and  
torque. This improves product quality, operational consistency, and process  
reliability in industrial applications.
```

```
[Source: ..\data\Adjustable Speed Drive.txt, score: 0.815]  
Ultimately, Adjustable Speed Drives are not merely devices for controlling  
motor speed; they are key technologies that support energy conservation,  
industrial innovation, and sustainable development in the modern world.
```

```
[Source: ..\data\Adjustable Speed Drive.txt, score: 0.807]  
An Adjustable Speed Drive is a system designed to vary the speed of an  
electric motor by controlling electrical input parameters such as voltage,  
frequency, or current. The primary purpose of an Adjustable Speed Drive is  
to match motor speed with the actual requirements of the process or machine  
being operated.
```

```
[Source: ..\data\Adjustable Speed Drive.txt, score: 0.798]  
An Adjustable Speed Drive is an electronic or electromechanical system used  
to control the speed, torque, and direction of an electric motor by varying  
the electrical power supplied to it. Adjustable Speed Drives are considered  
one of the most significant technological advancements in industrial  
automation because they provide flexibility, energy savings, smoother  
operation, and reduced mechanical stress on machinery. By adjusting motor  
speed according to load requirements, industries can optimize performance  
while minimizing electricity costs and equipment wear.
```

```
[Source: ..\data\Adjustable Speed Drive.txt, score: 0.790]  
The future of Adjustable Speed Drives is closely connected with the  
development of smart technologies, industrial automation, and sustainable  
energy systems. As industries seek more efficient and environmentally  
friendly solutions, Adjustable Speed Drives will continue to play a  
critical role in shaping the future of motor control and industrial  
operations.  
-----
```

The retrieval-validation judge drops two of the five as irrelevant, in full:

```
-----
[VALIDATE-RETRIEVAL] judging 5 chunk(s) for query: "What is an adjustable
speed drive and how does it control motor speed?"
-----

[VALIDATE-RETRIEVAL] verdict=PASS relevant=3/5
[VALIDATE-RETRIEVAL] reason: The majority of the chunks provide relevant
information about Adjustable Speed Drives and how they control motor speed,
making it plausible to support an answer.
    v chunk 0: Provides a general description of Adjustable Speed Drives.
    x chunk 1: Mentions the significance of Adjustable Speed Drives but
does not provide relevant information.
    v chunk 2: Describes the primary purpose of an Adjustable Speed Drive.
    v chunk 3: Provides a detailed explanation of how Adjustable Speed
Drives control motor speed.
    x chunk 4: Discusses the future of Adjustable Speed Drives but does not
provide relevant information for the question.
-----

[VALIDATE-RETRIEVAL] dropping 2 irrelevant chunk(s): [1, 4]
[VALIDATE-RETRIEVAL] surviving chunks (3):
    v orig_idx 0: Adjustable Speed Drives provide precise control over
motor speed and torque. This improves product q...
    v orig_idx 2: An Adjustable Speed Drive is a system designed to vary
the speed of an electric motor by controlling...
    v orig_idx 3: An Adjustable Speed Drive is an electronic or
electromechanical system used to control the speed, to...
[DEDUP] accumulated_chunks[1] ~ accumulated_chunks[2] (cos=0.962) --
will merge
    [1]: An Adjustable Speed Drive is a system designed to vary the speed
of an electric motor by controlling electrical input parameters such as
voltage, frequency, or current. The primary purpose of an Adjus...
    [2]: An Adjustable Speed Drive is an electronic or electromechanical
system used to control the speed, torque, and direction of an electric
motor by varying the electrical power supplied to it. Adjustable ...

[MERGE ATTEMPT 1/3] group keeper=accumulated_chunks[1], dups=[2]
```

And the complete merge — the LLM's raw JSON response, the merge validator's verdict, and the installation, all in full:

```
-----  
[CHUNK MERGE] consolidating 2 chunks from sources: ['..\data\Adjustable  
Speed Drive.txt', '..\data\Adjustable Speed Drive.txt']  
-----
```

```
[CHUNK MERGE] LLM response on attempt 1:  
{  
  "content": "An Adjustable Speed Drive is a system designed to vary the  
speed of an electric motor by controlling electrical input parameters such  
as voltage, frequency, or current [Source: ..\data\Adjustable Speed  
Drive.txt]. The primary purpose of an Adjustable Speed Drive is to match  
motor speed with the actual requirements of the process or machine being  
operated [Source: ..\data\Adjustable Speed Drive.txt]. An Adjustable  
Speed Drive is an electronic or electromechanical system used to control  
the speed, torque, and direction of an electric motor by varying the  
electrical power supplied to it [Source: ..\data\Adjustable Speed  
Drive.txt]. Adjustable Speed Drives are considered one of the most  
significant technological advancements in industrial automation because  
they provide flexibility, energy savings, smoother operation, and reduced  
mechanical stress on machinery [Source: ..\data\Adjustable Speed  
Drive.txt]. By adjusting motor speed according to load requirements,  
industries can optimize performance while minimizing electricity costs and  
equipment wear [Source: ..\data\Adjustable Speed Drive.txt].",  
  "sources": ["..\data\Adjustable Speed Drive.txt"],  
  "merged_from": 2  
}  
[CHUNK MERGE] success -- 1107 chars, 1 source(s)  
-----
```

```
-----  
[VALIDATE-MERGE] judging merge of 2 source chunk(s)  
-----
```

```
[VALIDATE-MERGE] verdict=FAITHFUL fabricated=0 dropped=0  
[VALIDATE-MERGE] reason: The MERGED CHUNK is faithful to the SOURCE  
CHUNKS, preserving all substantive facts and not inventing any new ones.  
-----
```

```
[MERGE INSTALLED] accumulated_chunks[1] replaced with merged chunk; dups  
[2] marked for drop. (verdict=FAITHFUL)  
[DEDUP] dropping 1 redundant chunk(s) from successful merges
```

```
[AGENT STATE] total_retrievals=1/5 | accumulated_chunks=2
```


B.4 Phase 2, call 2 of 3 — "motor speed control"

```
-----  
STEP: QUERY -> EMBEDDING  
-----  
Query       : "motor speed control"  
Model       : all-MiniLM-L6-v2  
Shape       : (384,)  
First 8 vals : [-0.0484,  0.043035, -0.038045,  0.023802, -0.040015,  
0.055872, -0.000218, -0.017619]  
L2 norm     : 1.000000 (1.0 = fully normalised)  
-----
```

```
Querying vector store -- top_k=5, score_threshold=0.5
```

```
Merged & ranked 5 doc(s) after dedup/threshold filter:
```

```
Doc 0 -- score: 0.6999 [..\data\Adjustable Speed Drive.txt]  
Doc 1 -- score: 0.6859 [..\data\Adjustable Speed Drive.txt]  
Doc 2 -- score: 0.6443 [..\data\Adjustable Speed Drive.txt]  
Doc 3 -- score: 0.6418 [..\data\Adjustable Speed Drive.txt]  
Doc 4 -- score: 0.6401 [..\data\Adjustable Speed Drive.txt]
```

```
-----  
RETRIEVED (unfiltered) -- retrieve_documents("motor speed control")  
5 chunk(s) | 2962 chars total -- pending relevance check  
-----
```

```
[Source: ..\data\Adjustable Speed Drive.txt, score: 0.700]
```

The concept of speed control is not entirely new. Earlier methods involved mechanical systems such as gearboxes, belt drives, and throttling valves. Although these methods provided some degree of speed variation, they were inefficient, bulky, and prone to wear and maintenance issues. Modern Adjustable Speed Drives replaced these mechanical approaches with electronic control systems that provide greater precision, reliability, and efficiency.

```
[Source: ..\data\Adjustable Speed Drive.txt, score: 0.686]
```

In modern industrial and commercial systems, the efficient control of electric motors has become one of the most important requirements for improving productivity, reducing energy consumption, and enhancing operational flexibility. Electric motors are widely used in manufacturing industries, transportation systems, water treatment plants, heating and ventilation systems, mining operations, robotics, and household appliances. Traditionally, motors operated at a constant speed determined by the frequency of the electrical supply and the design of the motor itself. However, many applications require motors to operate at varying speeds

depending on changing operational demands. This need led to the development of Adjustable Speed Drives (Adjustable Speed Drives), devices specifically designed to regulate and control motor speed efficiently and accurately.

[Source: ..\data\Adjustable Speed Drive.txt, score: 0.644]

The increasing emphasis on sustainability and energy conservation has further increased the importance of Adjustable Speed Drives in industrial and commercial sectors. According to energy studies, electric motors consume a significant portion of global electricity, especially in manufacturing facilities. By integrating Adjustable Speed Drives into motor-driven systems, organizations can reduce energy consumption substantially while improving process control and equipment lifespan. This essay explores the concept of Adjustable Speed Drives, including their principles, components, types, working mechanisms, applications, advantages, limitations, and future developments.

[Source: ..\data\Adjustable Speed Drive.txt, score: 0.642]

The future of Adjustable Speed Drives is closely connected with the development of smart technologies, industrial automation, and sustainable energy systems. As industries seek more efficient and environmentally friendly solutions, Adjustable Speed Drives will continue to play a critical role in shaping the future of motor control and industrial operations.

[Source: ..\data\Adjustable Speed Drive.txt, score: 0.640]

The motor is the device being controlled by the Adjustable Speed Drive. Adjustable Speed Drives can be used with various types of motors, including AC induction motors, synchronous motors, and DC motors. AC induction motors are the most common due to their durability, low maintenance, and widespread industrial use.

This variant loses three of five chunks to the relevance judge, and triggers no merge — the two survivors are not near-duplicates of anything already accumulated:

[VALIDATE-RETRIEVAL] judging 5 chunk(s) for query: "What is an adjustable speed drive and how does it control motor speed?"

[VALIDATE-RETRIEVAL] verdict=PASS relevant=2/5
[VALIDATE-RETRIEVAL] reason: Chunks 0 and 1 provide relevant information on the concept and development of Adjustable Speed Drives, which together could plausibly support an answer to the question.

v chunk 0: Mentions the replacement of mechanical systems with electronic control systems for speed variation.

v chunk 1: Describes the need for motor speed control in various applications and the development of Adjustable Speed Drives.

x chunk 2: Discusses the importance of Adjustable Speed Drives in terms of energy conservation, but does not provide information on

x chunk 3: Mentions the future of Adjustable Speed Drives, but does not provide information on how they control motor speed.

x chunk 4: Describes the types of motors that can be used with Adjustable Speed Drives, but does not provide information on how the

[VALIDATE-RETRIEVAL] dropping 3 irrelevant chunk(s): [2, 3, 4]

[VALIDATE-RETRIEVAL] surviving chunks (2):

v orig_idx 0: The concept of speed control is not entirely new. Earlier methods involved mechanical systems such a...

v orig_idx 1: In modern industrial and commercial systems, the efficient control of electric motors has become one...

[AGENT STATE] total_retrievals=2/5 | accumulated_chunks=4

B.5 Phase 2, call 3 of 3 — "adjustable speed drive motor control"

```
-----  
STEP: QUERY -> EMBEDDING  
-----
```

```
Query       : "adjustable speed drive motor control"  
Model       : all-MiniLM-L6-v2  
Shape       : (384,)  
First 8 vals : [-0.007812, 0.041869, 0.007568, 0.042462, -0.077237,  
0.084037, -0.052856, 0.001804]  
L2 norm     : 1.000000 (1.0 = fully normalised)  
-----
```

```
Querying vector store -- top_k=5, score_threshold=0.5
```

```
Merged & ranked 5 doc(s) after dedup/threshold filter:
```

```
Doc 0 -- score: 0.8321 [..\data\Adjustable Speed Drive.txt]  
Doc 1 -- score: 0.8140 [..\data\Adjustable Speed Drive.txt]  
Doc 2 -- score: 0.8042 [..\data\Adjustable Speed Drive.txt]  
Doc 3 -- score: 0.8016 [..\data\Adjustable Speed Drive.txt]  
Doc 4 -- score: 0.7988 [..\data\Adjustable Speed Drive.txt]
```

```
-----  
RETRIEVED (unfiltered) -- retrieve_documents("adjustable speed drive  
motor control")
```

```
5 chunk(s) | 2036 chars total -- pending relevance check  
-----
```

```
[Source: ..\data\Adjustable Speed Drive.txt, score: 0.832]  
Adjustable Speed Drives provide precise control over motor speed and  
torque. This improves product quality, operational consistency, and process  
reliability in industrial applications.
```

```
[Source: ..\data\Adjustable Speed Drive.txt, score: 0.814]  
Ultimately, Adjustable Speed Drives are not merely devices for controlling  
motor speed; they are key technologies that support energy conservation,  
industrial innovation, and sustainable development in the modern world.
```

```
[Source: ..\data\Adjustable Speed Drive.txt, score: 0.804]  
An Adjustable Speed Drive is an electronic or electromechanical system used  
to control the speed, torque, and direction of an electric motor by varying  
the electrical power supplied to it. Adjustable Speed Drives are considered  
one of the most significant technological advancements in industrial  
automation because they provide flexibility, energy savings, smoother  
operation, and reduced mechanical stress on machinery. By adjusting motor  
speed according to load requirements, industries can optimize performance
```

while minimizing electricity costs and equipment wear.

[Source: ..\data\Adjustable Speed Drive.txt, score: 0.802]

Adjustable Speed Drives represent one of the most important advancements in modern electrical and industrial engineering. By controlling the speed, torque, and direction of electric motors, Adjustable Speed Drives provide industries with improved efficiency, flexibility, and performance. Their ability to match motor operation with actual process requirements leads to substantial energy savings, reduced mechanical wear, and enhanced productivity.

[Source: ..\data\Adjustable Speed Drive.txt, score: 0.799]

An Adjustable Speed Drive is a system designed to vary the speed of an electric motor by controlling electrical input parameters such as voltage, frequency, or current. The primary purpose of an Adjustable Speed Drive is to match motor speed with the actual requirements of the process or machine being operated.

Four of five survive relevance filtering, and two independent merge groups form at once — the new chunks collide with both each other and with what earlier calls already installed:

[VALIDATE-RETRIEVAL] judging 5 chunk(s) for query: "What is an adjustable speed drive and how does it control motor speed?"

[VALIDATE-RETRIEVAL] verdict=PASS relevant=4/5

[VALIDATE-RETRIEVAL] reason: The retrieved chunks collectively provide a clear understanding of what an Adjustable Speed Drive is and how it controls motor speed, making them relevant to the original user question.

v chunk 0: Mentions the benefits of Adjustable Speed Drives

x chunk 1: Provides a general statement about Adjustable Speed Drives but does not contribute to answering the question

v chunk 2: Describes the function and benefits of Adjustable Speed Drives

v chunk 3: Highlights the importance and benefits of Adjustable Speed Drives

v chunk 4: Explains the primary purpose of an Adjustable Speed Drive

[VALIDATE-RETRIEVAL] dropping 1 irrelevant chunk(s): [1]

[VALIDATE-RETRIEVAL] surviving chunks (4):

v orig_idx 0: Adjustable Speed Drives provide precise control over motor speed and torque. This improves product q...

v orig_idx 2: An Adjustable Speed Drive is an electronic or electromechanical system used to control the speed, to...

v orig_idx 3: Adjustable Speed Drives represent one of the most

important advancements in modern electrical and in...

v orig_idx 4: An Adjustable Speed Drive is a system designed to vary the speed of an electric motor by controlling...

[DEDUP] accumulated_chunks[4] ~ accumulated_chunks[0] (cos=1.000) -- will merge

[4]: Adjustable Speed Drives provide precise control over motor speed and torque. This improves product quality, operational consistency, and process reliability in industrial applications....

[0]: Adjustable Speed Drives provide precise control over motor speed and torque. This improves product quality, operational consistency, and process reliability in industrial applications....

[DEDUP] accumulated_chunks[4] ~ accumulated_chunks[6] (cos=0.909) -- will merge

[4]: Adjustable Speed Drives provide precise control over motor speed and torque. This improves product quality, operational consistency, and process reliability in industrial applications....

[6]: Adjustable Speed Drives represent one of the most important advancements in modern electrical and industrial engineering. By controlling the speed, torque, and direction of electric motors, Adjustable...

[DEDUP] accumulated_chunks[5] ~ accumulated_chunks[1] (cos=0.936) -- will merge

[5]: An Adjustable Speed Drive is an electronic or electromechanical system used to control the speed, torque, and direction of an electric motor by varying the electrical power supplied to it. Adjustable ...

[1]: An Adjustable Speed Drive is a system designed to vary the speed of an electric motor by controlling electrical input parameters such as voltage, frequency, or current [Source: ..\data\Adjustable Spee...

[DEDUP] accumulated_chunks[5] ~ accumulated_chunks[7] (cos=0.962) -- will merge

[5]: An Adjustable Speed Drive is an electronic or electromechanical system used to control the speed, torque, and direction of an electric motor by varying the electrical power supplied to it. Adjustable ...

[7]: An Adjustable Speed Drive is a system designed to vary the speed of an electric motor by controlling electrical input parameters such as voltage, frequency, or current. The primary purpose of an Adjus...

[MERGE ATTEMPT 1/3] group keeper=accumulated_chunks[4], dups=[0, 6]

Both merges succeed and validate FAITHFUL, in full:

```
-----
[CHUNK MERGE] consolidating 3 chunks from sources: ['..\data\Adjustable
Speed      Drive.txt',      '..\data\Adjustable      Speed      Drive.txt',
'..\data\Adjustable Speed Drive.txt']
-----

[CHUNK MERGE] LLM response on attempt 1:
{"content": "Adjustable Speed Drives provide precise control over motor
speed and torque [Source: ..\data\Adjustable Speed Drive.txt]. This
improves product quality, operational consistency, and process reliability
in industrial applications [Source: ..\data\Adjustable Speed Drive.txt].
Adjustable Speed Drives represent one of the most important advancements in
modern electrical and industrial engineering [Source: ..\data\Adjustable
Speed Drive.txt]. By controlling the speed, torque, and direction of
electric motors, Adjustable Speed Drives provide industries with improved
efficiency, flexibility, and performance [Source: ..\data\Adjustable
Speed Drive.txt]. Their ability to match motor operation with actual
process requirements leads to substantial energy savings, reduced
mechanical wear, and enhanced productivity [Source: ..\data\Adjustable
Speed Drive.txt].", "sources": ["..\data\Adjustable Speed Drive.txt"],
"merged_from": 3}
[CHUNK MERGE] success -- 859 chars, 1 source(s)
-----

-----
[VALIDATE-MERGE] judging merge of 3 source chunk(s)
-----

[VALIDATE-MERGE] verdict=FAITHFUL fabricated=0 dropped=0
[VALIDATE-MERGE] reason: The MERGED CHUNK is faithful to the SOURCE
CHUNKS, preserving all substantive facts and not inventing any new ones.
-----

[MERGE INSTALLED] accumulated_chunks[4] replaced with merged chunk; dups
[0, 6] marked for drop. (verdict=FAITHFUL)

[MERGE ATTEMPT 1/3] group keeper=accumulated_chunks[5], dups=[1, 7]
-----

[CHUNK MERGE] consolidating 3 chunks from sources: ['..\data\Adjustable
Speed      Drive.txt',      '..\data\Adjustable      Speed      Drive.txt',
'..\data\Adjustable Speed Drive.txt']
-----

[CHUNK MERGE] LLM response on attempt 1:
{"content": "An Adjustable Speed Drive is an electronic or
electromechanical system used to control the speed, torque, and direction
of an electric motor by varying the electrical power supplied to it
[Source: ..\data\Adjustable Speed Drive.txt]. The primary purpose of an
```

```

Adjustable Speed Drive is to match motor speed with the actual requirements
of the process or machine being operated [Source: ..\\data\\Adjustable
Speed Drive.txt]. Adjustable Speed Drives are considered one of the most
significant technological advancements in industrial automation because
they provide flexibility, energy savings, smoother operation, and reduced
mechanical stress on machinery [Source: ..\\data\\Adjustable Speed
Drive.txt]. By adjusting motor speed according to load requirements,
industries can optimize performance while minimizing electricity costs and
equipment wear [Source: ..\\data\\Adjustable Speed Drive.txt].", "sources":
["..\\data\\Adjustable Speed Drive.txt"], "merged_from": 3}
[CHUNK MERGE] success -- 893 chars, 1 source(s)
-----

[VALIDATE-MERGE] judging merge of 3 source chunk(s)
-----

[VALIDATE-MERGE] verdict=FAITHFUL fabricated=0 dropped=0
[VALIDATE-MERGE] reason: The MERGED CHUNK is faithful to the SOURCE
CHUNKS, preserving all substantive facts and not inventing any new ones.
-----

[MERGE INSTALLED] accumulated_chunks[5] replaced with merged chunk; dups
[1, 7] marked for drop. (verdict=FAITHFUL)
[DEDUP] dropping 4 redundant chunk(s) from successful merges

[AGENT STATE] total_retrievals=3/5 | accumulated_chunks=4

```

Three of the run's five allowed retrievals are now spent, because the LLM issued all three tool calls in a single turn rather than one at a time — the fan-out this appendix's B.2 response already showed.

B.6 Phase 3 — the quality gate

Iteration 2's message payload re-sends the entire conversation so far — the identical system prompt, question, and all three tool results already quoted complete in B.3–B.5 above — which is why cumulative prompt tokens climb rather than reset between turns:

```
[CTXSIZE]      iter=2      msgs=6      chars=6389      prompt_tokens=1724
cum_prompt_tokens=2544
```

Iteration 2's response is a single tool call, `check\_answer\_quality`, carrying the complete draft answer as its argument, in full:

```
tool_calls requested (1):
-> name : check_answer_quality
   id   : mg9re96py
   args : {
     "answer": "An Adjustable Speed Drive is a system designed to vary the
speed of an electric motor by controlling electrical input parameters such
as voltage, frequency, or current. The primary purpose of an Adjustable
Speed Drive is to match motor speed with the actual requirements of the
process or machine being operated. Adjustable Speed Drives provide precise
control over motor speed and torque. This improves product quality,
operational consistency, and process reliability in industrial
applications. An Adjustable Speed Drive is an electronic or
electromechanical system used to control the speed, torque, and direction
of an electric motor by varying the electrical power supplied to it.
Adjustable Speed Drives are considered one of the most significant
technological advancements in industrial automation because they provide
flexibility, energy savings, smoother operation, and reduced mechanical
stress on machinery. By adjusting motor speed according to load
requirements, industries can optimize performance while minimizing
electricity costs and equipment wear."
   }
}
```

```
[AGENT] injecting 3246 chars of accumulated context (4 chunk(s)) into
check_answer_quality
[AGENT STATE] final_quality set to: OK

[7] role=tool tool_call_id=mg9re96py
   content: OK -- now write your final answer. Expand it using all
retrieved chunks. Cite each source inline as [Source: filename].
```

B.7 Phase 4 — the final answer is drafted

```
[CTXSIZE]          iter=3          msgs=8          chars=6507          prompt_tokens=1947
cum_prompt_tokens=4491
```

Iteration 3's response has no tool calls — the orchestrator's own log line marks the turn explicitly, `'tool_calls: none -> this is the FINAL ANSWER'`. The CLI's own live preview of this response truncates it mid-sentence at 2,858 characters (“[Source: Adjustable Spee...”); its complete, untruncated text is identical to what prints under `'FINAL ANSWER'` after compression runs, quoted whole in B.9 below, since compression in this build operates on the retrieved chunks, not on the drafted answer text itself.

```
=====
LLM RESPONSE  [Iteration 3]
=====
Tokens this call : 1947 prompt + 476 completion = 2423 total
content (2858 chars):
    An Adjustable Speed Drive is a system designed to vary the speed of an
    electric motor by controlling electrical input parameters such as voltage,
    frequency, or current. The primary purpose of an Adjustable Speed Drive is
    to match motor speed with the actual requirements of the process or machine
    being operated [Source: Adjustable Speed Drive.txt]. This is particularly
    important in industrial applications where precise control over motor speed
    and torque is crucial for improving product quality, operational
    consistency, and process reliability [Source: Adjustable Speed Drive.txt].

    An Adjustable Speed Drive is an electronic or electromechanical system used
    to control the speed, torque, and direction of an electric motor by varying
    the electrical power supplied to it [Source: Adjustable Spee...

    tool_calls: none -> this is the FINAL ANSWER
=====
```

B.8 Phase 5 — compression: NAC, DC, and a guard that overrides the LLM

Compression in this build runs on the way to the printed final answer, logged under its own explicit banner. NAC's complete merge, in full:

```
[NAC] Running Neighbor-Aware Compression before final answer...

-----

[NAC] Neighbor-Aware Compression -- scanning 4 chunk(s)

-----

[NAC] 1 neighbor run(s) found | 2 singleton(s) pass through unchanged
run: source=..\data\Adjustable Speed Drive.txt seq=2->3 (2 chunks)

[NAC MERGE attempt 1/3] 2 chunks from '..\data\Adjustable Speed
Drive.txt' seq 2->3

-----

[CHUNK MERGE] consolidating 2 chunks from sources: ['..\data\Adjustable
Speed Drive.txt', '..\data\Adjustable Speed Drive.txt']

-----

[CHUNK MERGE] LLM response on attempt 1:
{"content": "In modern industrial and commercial systems, the efficient
control of electric motors has become one of the most important
requirements for improving productivity, reducing energy consumption, and
enhancing operational flexibility. Electric motors are widely used in
manufacturing industries, transportation systems, water treatment plants,
heating and ventilation systems, mining operations, robotics, and household
appliances. Traditionally, motors operated at a constant speed determined
by the frequency of the electrical supply and the design of the motor
itself. However, many applications require motors to operate at varying
speeds depending on changing operational demands. This need led to the
development of Adjustable Speed Drives (Adjustable Speed Drives), devices
specifically designed to regulate and control motor speed efficiently and
accurately. An Adjustable Speed Drive is an electronic or electromechanical
system used to control the speed, torque, and direction of an electric
motor by varying the electrical power supplied to it [Source:
..\data\Adjustable Speed Drive.txt]. The primary purpose of an Adjustable
Speed Drive is to match motor speed with the actual requirements of the
process or machine being operated [Source: ..\data\Adjustable Speed
Drive.txt]. Adjustable Speed Drives are considered one of the most
significant technological advancements in industrial automation because
they provide flexibility, energy savings, smoother operation, and reduced
mechanical stress on machinery [Source: ..\data\Adjustable Speed
Drive.txt]. By adjusting motor speed according to load requirements,
industries can optimize performance while minimizing electricity costs and
```

```

equipment wear [Source: ..\\data\\Adjustable Speed Drive.txt].", "sources":
["..\\data\\Adjustable Speed Drive.txt"], "merged_from": 2}
[CHUNK MERGE] success -- 1757 chars, 1 source(s)
-----

[VALIDATE-MERGE] judging merge of 2 source chunk(s)
-----

[VALIDATE-MERGE] verdict=FAITHFUL fabricated=0 dropped=0
[VALIDATE-MERGE] reason: The MERGED CHUNK is faithful to the SOURCE
CHUNKS, preserving all substantive facts and not inventing any new ones.
-----

[NAC MERGE] success on attempt 1 (verdict=FAITHFUL)

[NAC] compression done: 4 -> 3 chunk(s) (1 eliminated)
-----

```

DC's complete redundancy scan across the resulting three chunks, all three candidate groups and both verdicts, in full:

```

[DC] Running Deduplication Compression after NAC...
-----

[DC] Deduplication Compression -- 3 chunk(s), window=3
-----

[DC] scanning window [0:3] (3 chunk(s))
[DC] 3 redundancy group(s) identified in this window:
    chunk_index=0 sentence="An Adjustable Speed Drive is an electronic or
electromechanical system used to control the speed, torque, and direction "
    chunk_index=2 sentence="By controlling the speed, torque, and
direction of electric motors, Adjustable Speed Drives provide industries
with impr"
    chunk_index=0 sentence="Adjustable Speed Drives are considered one of
the most significant technological advancements in industrial automation b"
    chunk_index=2 sentence="Adjustable Speed Drives represent one of the
most important advancements in modern electrical and industrial
engineering"
    chunk_index=1 sentence="Modern Adjustable Speed Drives replaced these
mechanical approaches with electronic control systems that provide greater"
    chunk_index=2 sentence="Adjustable Speed Drives provide precise
control over motor speed and torque."
-----

[VALIDATE-REDUNDANCY] judging 3 group(s) across 3 chunk(s)
-----

```

```

[VALIDATE-REDUNDANCY] recovered JSON via first/last bracket extraction (3
item(s))
[VALIDATE-REDUNDANCY] confirmed=2 rejected=1
  v group 0 CONFIRMED: Both sentences refer to the same technology class
(ASD) and state the same core fact (precise control of motor speed/tor
    chunk 0: "An Adjustable Speed Drive is an electronic or
electromechanical system used to control the speed, to"
    chunk 2: "By controlling the speed, torque, and direction of
electric motors, Adjustable Speed Drives provide "
  v group 2 CONFIRMED: Both sentences refer to the same technology class
(ASD) and state the same core fact (precise control of motor speed and
    chunk 1: "Modern Adjustable Speed Drives replaced these mechanical
approaches with electronic control systems "
    chunk 2: "Adjustable Speed Drives provide precise control over
motor speed and torque."
  x group 1 REJECTED: The sentences refer to the same technology class
(ASD), but one sentence mentions 'flexibility, energy savings, smoother
    chunk 0: "Adjustable Speed Drives are considered one of the most
significant technological advancements in ind"
    chunk 2: "Adjustable Speed Drives represent one of the most
important advancements in modern electrical and in"
-----
[DC] chunk 2: no change after removal attempt -- skipping

[DC] done -- 0 redundant passage(s) removed across 3 chunk(s)
-----

```

Two of DC's three candidate groups are confirmed redundant by the judge, yet zero passages are actually removed — a real, observed case of a confirmed-redundant group not collapsing into anything, worth knowing as a genuine DC fragility rather than assuming a confirmed verdict always changes the output.

LBC then runs per chunk, and this trace catches its over-compression guard in the act — twice — with every reason string complete:

```

[LBC] Running LLM-Based Compression after DC...

-----

[LBC] LLM-Based Compression -- 3 chunk(s)
[LBC] query: "What is an adjustable speed drive and how does it control
motor speed?"
-----

[LBC] chunk 0 -- source: '..\data\Adjustable Speed Drive.txt' (1757
chars)
[LBC] chunk 0: LLM produced 332 chars (dropped_count=6) reason: Removed
sentences that describe document structure rather than facts, and sentences
that repeat context already present

```

```
[LBC] chunk 0: retention ratio 0.19 < 0.35 -- keeping original (over-compression guard)
```

```
[LBC] chunk 1 -- source: '..\data\Adjustable Speed Drive.txt' (445 chars)
```

```
[LBC] chunk 1: LLM produced 160 chars (dropped_count=3) reason: The original chunk contained too much background information and metadata, making it necessary to drop most of the conte
```

```
-----  
[VALIDATE-LBC] judging compressed chunk from source: '..\data\Adjustable Speed Drive.txt'
```

```
[VALIDATE-LBC] query: "What is an adjustable speed drive and how does it control motor speed?"
```

```
-----  
[VALIDATE-LBC] verdict=SAFE fabricated=0 lost_relevant=0
```

```
[VALIDATE-LBC] reason: No fabrications or lost relevant facts found.
```

```
-----  
[LBC] chunk 1: SAFE -- installing (445 -> 160 chars, -285 chars)
```

```
[LBC] chunk 2 -- source: '..\data\Adjustable Speed Drive.txt' (859 chars)
```

```
[LBC] chunk 2: LLM produced 166 chars (dropped_count=5) reason: Only one sentence directly helps answer the query about Adjustable Speed Drives and how they control motor speed.
```

```
[LBC] chunk 2: retention ratio 0.19 < 0.35 -- keeping original (over-compression guard)
```

```
[LBC] done -- 3 -> 3 chunk(s) (0 dropped as irrelevant, 1 compressed, 2 kept as-is)
```

```
[LBC] chars: 3,061 -> 2,776 (-285 = 9.3% reduction)  
-----
```

Chunks 0 and 2 both fall below the 0.35 retention floor and are kept verbatim, overriding the LLM's own compression rather than trusting it; only chunk 1's more conservative compression clears the bar and is validated 'SAFE' before installation. The net result — a modest 9.3% reduction — is the honest number for what the guards produce when they are actively rejecting aggressive compression, not a dramatic space saving.

B.9 The final answer and the true session cost

The complete printed answer — all four paragraphs, every inline citation, exactly as the CLI printed it:

```
=====
FINAL ANSWER
=====

An Adjustable Speed Drive is a system designed to vary the speed of an
electric motor by controlling electrical input parameters such as voltage,
frequency, or current. The primary purpose of an Adjustable Speed Drive is
to match motor speed with the actual requirements of the process or machine
being operated [Source: Adjustable Speed Drive.txt]. This is particularly
important in industrial applications where precise control over motor speed
and torque is crucial for improving product quality, operational
consistency, and process reliability [Source: Adjustable Speed Drive.txt].

An Adjustable Speed Drive is an electronic or electromechanical system used
to control the speed, torque, and direction of an electric motor by varying
the electrical power supplied to it [Source: Adjustable Speed Drive.txt].
This technology has been considered one of the most significant
advancements in industrial automation because it provides flexibility,
energy savings, smoother operation, and reduced mechanical stress on
machinery [Source: Adjustable Speed Drive.txt]. By adjusting motor speed
according to load requirements, industries can optimize performance while
minimizing electricity costs and equipment wear [Source: Adjustable Speed
Drive.txt].

The concept of speed control is not entirely new. Earlier methods involved
mechanical systems such as gearboxes, belt drives, and throttling valves
[Source: Adjustable Speed Drive.txt]. Although these methods provided some
degree of speed variation, they were inefficient, bulky, and prone to wear
and maintenance issues [Source: Adjustable Speed Drive.txt]. Modern
Adjustable Speed Drives replaced these mechanical approaches with
electronic control systems that provide greater precision, reliability, and
efficiency [Source: Adjustable Speed Drive.txt].

In modern industrial and commercial systems, the efficient control of
electric motors has become one of the most important requirements for
improving productivity, reducing energy consumption, and enhancing
operational flexibility [Source: Adjustable Speed Drive.txt]. Electric
motors are widely used in manufacturing industries, transportation systems,
water treatment plants, heating and ventilation systems, mining operations,
robotics, and household appliances [Source: Adjustable Speed Drive.txt].
Traditionally, motors operated at a constant speed determined by the
frequency of the electrical supply and the design of the motor itself
[Source: Adjustable Speed Drive.txt]. However, many applications require
motors to operate at varying speeds depending on changing operational
```

demands [Source: Adjustable Speed Drive.txt]. This need led to the development of Adjustable Speed Drives (Adjustable Speed Drives), devices specifically designed to regulate and control motor speed efficiently and accurately [Source: Adjustable Speed Drive.txt].

```
=====
Iterations used : 3 / 6
Quality signal  : OK
Tokens (total)  : 4491 prompt + 1241 completion = 5732 total
Sources searched:
  * "adjustable speed drive"
    preview: [Source: ..\data\Adjustable Speed Drive.txt, score: 0.826]
Adjustable Speed Drives provide precise c...
  * "motor speed control"
    preview: [Source: ..\data\Adjustable Speed Drive.txt, score: 0.700]
The concept of speed control is not entir...
  * "adjustable speed drive motor control"
    preview: [Source: ..\data\Adjustable Speed Drive.txt, score: 0.832]
Adjustable Speed Drives provide precise c...
=====
Your question:
```

Every claim in the printed answer carries an inline `[Source: ...]` citation, and the footer's 5,732-token total ties back exactly to the three `[CTXSIZE]` lines logged along the way — 820, then 2,544 cumulative prompt tokens, then 4,491, plus 1,241 completion tokens summed across the three LLM calls. Three of the run's six allowed iterations and three of its five allowed retrievals were spent; nothing in this run came close to either cap. The trailing `Your question:` is the CLI's own prompt for the next turn — this session ends here with no follow-up typed.

B.10 Where Phoenix and LangSmith span IDs actually appear

As B.1 flags, no span ID belongs inside the trace above. What follows is a separate, later, differently-sourced excerpt — real LangSmith trace/run UUIDs from ``Runs/langchain_api_20260710_115823.debug.log``, an ``app/``-side request captured after ADR-063/064 shipped, and a Phoenix/Langfuse OTEL collector confirmation from a same-era ``app_workflow/`` log:

```
2026-07-10 11:59:04 | DEBUG | langsmith.client | [-] | Sending compressed
multipart request with context:
trace=019f4ad2-9892-73d0-b7f7-cb5d9f239857,id=019f4ad2-9892-73d0-b7f7-
cb5d9f239857;
trace=019f4ad2-eadd-74e0-86d8-52e937afe168,id=019f4ad2-eadd-74e0-86d8-
52e937afe168

2026-07-14 17:23:54 | DEBUG | urllib3.connectionpool | [-] |
http://localhost:3000 "POST /api/public/otel/v1/traces HTTP/1.1" 200 None
```

These are wire-protocol debug lines, not the narrative ``[CTXSIZE]`/[NAC]`/[LBC]`` logging this appendix has walked through — Decisions.md's ADR-064 records that span `*events*` in this system are looked up programmatically (``LangChainInstrumentor()._tracer.get_span(run_id)``, ``Client().read_run(run_id)``) rather than printed inline the way context-size telemetry is, so even a fresh dry run of the instrumented ``app_workflow/`` pipeline would not print a span ID next to the phase that produced it the way ``[CTXSIZE]`` does. Chapter 38 covers how those spans are actually queried and viewed.